

$$z = x + iy = r e^{i\theta}$$

$$r = \sqrt{x^2 + y^2} \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$f(z) = u(x, y) + i v(x, y)$$

$$\begin{aligned} f(z) &= \frac{1}{z} = \frac{1}{x+iy} \\ &= \frac{1}{x+iy} \cdot \frac{x-iy}{x-iy} \\ &= \frac{x-iy}{x^2+y^2} \\ &= \underbrace{\frac{x}{x^2+y^2}}_u - i \underbrace{\frac{y}{x^2+y^2}}_v \end{aligned}$$

$$f(z) = \log(z) = \log(r e^{i\theta})$$

$$= \log(r) + \log(e^{i\theta})$$

$$f(z) = \log(r) + i\theta$$

$$= \log\sqrt{x^2+y^2} + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$f(z) = \underbrace{\frac{1}{2} \log(x^2+y^2)}_u + i \underbrace{\tan^{-1}\left(\frac{y}{x}\right)}_v$$

(C-R eqn -)

$$f(z) = u(x, y) + i v(x, y)$$

$$\frac{\partial u}{\partial x}$$

$$\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial y}$$

$$\frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$f(z) = u(x, y) + i v(x, y)$$

Analytic function
الدالة التحليلية

u, v continuous

① $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ continuous in domain D

② u, v satisfy the C-R equation at all point of D

Entire function
الدالة الشاملة (الكاملة)

A function which is analytic every where
إذا كانت تحليلية في كل مكان (جميع نقاط المستوى المركب)

$$f(z) = z^2 - 3z + 1$$

$$f(z) = e^z$$

$$f(z) = \sin(z)$$

$$f(z) = \cos(z)$$

Singular point
النقطة الشاذة

The point at which the function is not differentiable

$$f(z) = \frac{1}{z}$$

$$f(z) = \frac{1}{z}$$

$z=0 \rightarrow$ singular point

(C-R eqn.)

$$f(z) = u(x, y) + i v(x, y)$$

$$\frac{\partial u}{\partial x}$$

$$\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial y}$$

$$\frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$f(z) = u(x, y) + i v(x, y)$$

u, v continuous

Analytic function
الدالة تحليلية

① $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ continuous in domain D

② u, v satisfy the C-R equation at all point of D

Entire function
الدالة الشاملة (الكليّة)

A function which is analytic every where
إذا كانت تحليلية في كل مكان (جميع نقاط المستوى المركب)

$$f(z) = z^2 - 3z + 1$$

$$f(z) = e^z$$

$$f(z) = \sin(z)$$

$$f(z) = \cos(z)$$

singular point
النقطة الشاذة

The point at which
the function is not
differentiable

$$f(z) = \frac{1}{z}$$

$z=0 \rightarrow$ singular point

$$f(z) = e^{\frac{1}{z}}$$

$$f(z) = w$$

$$\frac{dz}{dw} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

if $f(z)$ is analytic
 $f(z) = u + iv$

$$\bar{f}(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

or

$$\bar{f}(z) = \frac{\partial u}{\partial y} - i \frac{\partial v}{\partial y}$$

$$\begin{aligned} \sin(-x) &= -\sin x \\ \cos(-x) &= \cos x \end{aligned}$$

$$\sin(ix) = i \sinh(x)$$

$$\cos(ix) = \cosh(x)$$

$$\sinh(ix) = i \sin(x)$$

$$\cosh(ix) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \quad \frac{1}{i} = -i$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\sin(ix) = \frac{e^{i(ix)} - e^{-i(ix)}}{2i}$$

$$= \frac{e^{-x} - e^x}{2i}$$

$$= (-i) \frac{e^x - e^{-x}}{2}$$

$$= i \frac{e^x - e^{-x}}{2} = i \sinh(x)$$

verify that the function $f(z)$

is analytic
تحليلية

$$f(z) = e^z$$

sol

$$f(z) = e^{x+iy}$$

$$= e^x \cdot e^{iy}$$

$$= e^x [\cos(y) + i \sin(y)]$$

$$= \underbrace{e^x \cos(y)}_u + i \underbrace{e^x \sin(y)}_v$$

$$\frac{\partial u}{\partial x} = e^x \cos(y)$$

$$\frac{\partial v}{\partial x} = e^x \sin(y)$$

$$\frac{\partial u}{\partial y} = -e^x \sin(y)$$

$$\frac{\partial v}{\partial y} = e^x \cos(y)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$$

continuous
in every where
استمرارية في كل مكان

تحليلية
 $f(z)$ analytic
and entire
كلية

Verify that the function $f(z)$

is analytic
تحليلية

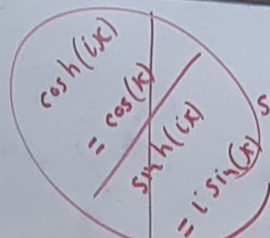
$$f(z) = \cosh(z)$$

sol

$$f(z) = \cosh(x+iy)$$

$$= \cosh(x) \cosh(iy) + \sinh(x) \sinh(iy)$$

$$= \underbrace{\cosh(x) \cos(y)}_u + i \underbrace{\sinh(x) \sin(y)}_v$$



$$\sinh(z) = \sinh(x+iy)$$

$$= \sinh(x) \cosh(iy) + \sinh(iy) \cosh(x)$$

$$= \underbrace{\sinh(x) \cos(y)}_u + i \underbrace{\sin(y) \cosh(x)}_v$$

$$\sin(z) = \sin(x+iy)$$

$$= \sin(x) \cos(iy) + \sin(iy) \cos(x)$$

$$= \sin(x) \cosh(y) + i \sinh(y) \cos(x)$$

$$\begin{aligned} \cos(z) &= \cos(x+iy) = \cos(x) \cos(iy) - \sin(x) \sin(iy) \\ &= \cos(x) \cosh(y) - i \sin(x) \sinh(y) \end{aligned}$$

verify that the function $f(z)$

is entire

$$f(z) = \cosh(z)$$

sol

$$f(z) = \cosh(x+iy)$$

$$= \cosh(x) \cosh(iy) + \sinh(x) \sinh(iy)$$

$$= \underbrace{\cosh(x) \cos(y)}_u + i \underbrace{\sinh(x) \sin(y)}_v$$

$$u = \cosh(x) \cos(y)$$

$$v = \sinh(x) \sin(y)$$

$$\frac{\partial u}{\partial x} = \sinh(x) \cos(y)$$

$$\frac{\partial v}{\partial x} = \cosh(x) \sin(y)$$

$$\frac{\partial u}{\partial y} = -\cosh(x) \sin(y)$$

$$\frac{\partial v}{\partial y} = \sinh(x) \cos(y)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \text{ continuous}$$

$f(z)$ analytic in every where
and $f(z)$ entire

If $w = \log(z)$ find $\frac{dw}{dz}$
and determine where w is non-analytic

sol

$$w = \log(r) + i\theta$$

$$w = \log \sqrt{x^2 + y^2} + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$w = \underbrace{\frac{1}{2} \log(x^2 + y^2)}_u + i \underbrace{\tan^{-1}\left(\frac{y}{x}\right)}_v$$

$$\frac{\partial u}{\partial x} = \frac{1}{x} \cdot \frac{2x}{x^2 + y^2} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial v}{\partial x} = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{x} \cdot \frac{2y}{x^2 + y^2} = \frac{y}{x^2 + y^2}$$

$$\frac{\partial v}{\partial y} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial v}{\partial x} = \frac{\left(\frac{y}{x}\right)'}{1 + \left(\frac{y}{x}\right)^2} = \frac{-\frac{y}{x^2}}{1 + \frac{y^2}{x^2}} = \frac{-y}{x^2 + y^2}$$

$$\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$$

$$\left(\frac{y}{x}\right)' = y \left(\frac{1}{x}\right)'$$

$$\frac{\partial v}{\partial y} = \frac{\left(\frac{y}{x}\right)'}{1 + \left(\frac{y}{x}\right)^2} = \frac{\frac{1}{x}}{1 + \frac{y^2}{x^2}} = \frac{x}{x^2 + y^2}$$

$$= y \left(\frac{-1}{x^2}\right)$$

$$= -\frac{y}{x^2}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$f(z)$ continuous except at $x=0$
 $y=0$

$f(z)$ analytic except at $z=0$

$\therefore$ at $z=0$ $f(z)$ non-analytic

$$\frac{dw}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{x}{x^2 + y^2} + i \frac{-y}{x^2 + y^2} = \frac{x - iy}{x^2 + y^2} = \frac{x - iy}{(x - iy)(x + iy)} = \frac{1}{x + iy} = \frac{1}{z}$$

$$\left(\frac{x}{z}\right)' = \frac{1}{z}$$

$$\left(\frac{x}{y}\right)' = \frac{1}{y}$$